Causal sufficient dimension reduction for studying environmental mixtures

An applied introduction and tutorial for the `csdr` R package

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Why CSDR?

What do we want to learn from an environmental mixture?

Scientific goals

- Estimate the **joint health effect**
- Learn the **shape** of the response
- Identify **important contributors**
- Find a useful **mixture summary**

The recurring problem

Highly correlated exposures, sparse support, and confounding—all at once.

$$(X_1, \ldots, X_p) \longrightarrow Y$$

Different methods prioritize different scientific goals.

Mixtures methods make different tradeoffs

Method	Dimension	What is learned?	Natural strength
WQS / qgcomp	$d = 1$	Scalar joint shift	Interpretable effect
PCA / PCP	$1 < d < p$	Exposure variation	Pattern discovery
BKMR / BART	$d = p$	Full exposure-response	Nonlinear interactions
CSDR	**learn** d	**Causal response directions**	**Flexible causal summary**

Key distinction

CSDR learns the structure in exposure X needed to preserve the **causal** exposure-response.

Why not always reduce to a scalar?

Why not always estimate the full exposure-response?

One exposure

Many observations support a curve.

$$\mu(x_1)$$

Two exposures

Observations thin across a surface.

$$\mu(x_1, x_2)$$

Many exposures

Most joint exposure combinations are sparse.

$$\mu(x_1, \ldots, x_p)$$

Consequence

More dimensions mean harder nuisance estimation, weaker support, and a response surface that is difficult to communicate.

CSDR is similar to a causally supervised PCA

$$\mu(x) = \mathbb{E}(Y^x) = g(\beta^\top x), \qquad \beta \in \mathbb{R}^{p\times d}, \qquad d \ll p.$$

Exposures

$$X_1,\ X_2,\ \dots,\ X_p$$

Causal summary

$$Z = \beta^\top X \in \mathbb{R}^d$$

Flexible response

$$\mu_\beta(z) = g(z)$$

Plain-language target

Replace p exposures with d summaries **without losing the causal mean response**.

Causal directions can differ from associational directions

Observed regression

$$\mathbb{E}(Y \mid X = x)$$

Confounding can create or obscure exposure directions associated with the outcome.

$$X \longleftarrow C \longrightarrow Y$$

Causal response

$$\mathbb{E}(Y^x)$$

CSDR seeks directions that preserve the response under interventions on X.

$$X \longrightarrow Y$$

The distinction that matters

Association-preserving $\neq$ causal-effect-preserving

A modular two-stage strategy

1. Adjust

Use (Y, X, C) to construct a cross-fitted causal response $\widetilde{Y}$.

$$\mathbb{E}(\widetilde{Y} \mid X) = \mu(X)$$

2. Reduce

Apply sufficient dimension reduction to $(\widetilde{Y}, X)$.

$$(\hat{\beta}, \hat{d})$$

3. Estimate

Project exposures and estimate the causal response in d dimensions.

$$\widehat{Z} = \hat{\beta}^\top X$$

Causal nuisance estimation $\longrightarrow$ standard SDR $\longrightarrow$ reduced causal ERS

What does the researcher get?

Dimension

$$\hat{d}$$

How many causal directions are needed?

Importance

$$\hat{\lambda}_j$$

Which exposures contribute to the subspace?

Response

$$\hat{\mu}(z)$$

What is the shape of the joint causal response?

Subspace importance score

$$\lambda_j = P_{jj}, \qquad P = \beta(\beta^\top \beta)^{-1}\beta^\top, \qquad \sum_{j=1}^{p} \lambda_j = d.$$

From method to evidence

Reduction can improve the final scientific estimate

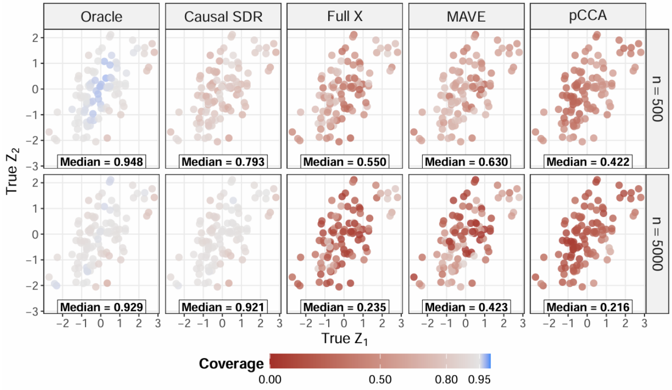


Figure 1: Pointwise coverage for the causal exposure–response surface

Simulation message: CSDR improved accuracy and coverage versus full X and noncausal reductions.

PFAS application: four exposures summarized in one direction

Study: 305 term births in the Atlanta African American Maternal-Child Cohort (ATL-AA)

Exposures: PFOS, PFOA, PFNA, PFHxS

Selected dimension: $\hat{d} = 1$

$$\widehat{Z} = 0.77\,PFOS + 0.14\,PFOA + 0.26\,PFNA - 0.56\,PFHxS$$

SIS: PFOS 0.60; PFOA 0.02; PFNA 0.07; PFHxS 0.31

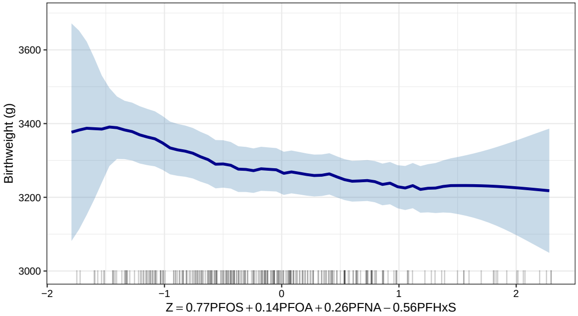


Figure 2: PFAS exposure-response curve from ATL-AA

Where CSDR fits in the analyst's toolbox

If the scientific question is...	A natural starting point is...
Effect of increasing all exposures by one quantile	qgcomp
Flexible response in the original p-dimensional space	BKMR / BART
Shared exposure patterns, sources, or unusual events	PCP / `pcpr`
Lowest-dimensional summary preserving the causal mean	**CSDR** / **`csdr`**

CSDR is especially attractive when

A low-dimensional causal structure is scientifically plausible, but neither $d = 1$ nor $d = p$ should be imposed in advance.

The `csdr` workflow

Every analysis starts with three objects

```r
library(csdr)

Y <- dat$birth_weight

A_vars <- c("PFOS", "PFOA", "PFNA", "PFHxS")
A <- dat[A_vars]

C <- model.matrix(
  ~ age + bmi + education + tobacco +
    marijuana + sex,
  data = dat
)[, -1, drop = FALSE]
```

Outcome

Y: health endpoint

Exposures

A: continuous mixture

Confounders

C: sufficient adjustment set

Notation bridge: The package calls the exposure mixture `A`; this is the X used in the causal framework.

Fit the causal subspace

```r
learners <- csdr_learners(sl_library = "SL.glm")

fit <- csdr(
  Y = Y, A = A, C = C,
  variants = "DR",
  d = NULL, max_dim = 3,
  L = 5, seed = 42,
  learners = learners,
  keep_nuisance = TRUE,
  verbose = FALSE
)
```

What happens

1. Cross-fit nuisance models
2. Construct $\widetilde{Y}$
3. Run MAVE
4. Select $\hat{d}$
5. Return the fitted subspace

DR is the package default. `SL.glm` keeps the tutorial fast; use a prespecified richer library for the substantive analysis.

Inspect dimension and importance before effects

```
summary(fit)

B_hat <- coef(fit, variant = "DR")
Z_hat <- scores(fit, variant = "DR")

mave <- mave_fits(fit, variant = "DR")
dim_selection <- mave$dimension_selection

Q_hat <- qr.Q(qr(B_hat))
sis <- rowSums(Q_hat^2)
```

PFAS SIS: PFOS 0.60; PFOA 0.02; PFNA 0.07; PFHxS 0.31

Visual placeholder

Replace with package output

Dimension CV curve

+

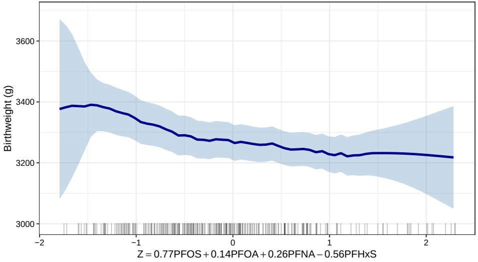
SIS dot plot with bootstrap intervals

Estimate the response in the reduced exposure

```r
z_grid <- matrix(seq(
  quantile(Z_hat[, 1], 0.05),
  quantile(Z_hat[, 1], 0.95),
  length.out = 50
), ncol = 1)

ers <- estimate_ers(
  Y = Y, A = Z_hat, C = C,
  a_eval = z_grid, estimator = "DR",
  L = 5, seed = 42,
  verbose = FALSE
)

ers$results
```



Read the rug before the curve: Uncertainty and causal interpretation are strongest where the reduced exposure has adequate support.

A credible analysis includes diagnostics

Check

- Dimension-selection profile
- Distribution and support of $\widehat{Z}$
- Nuisance-model performance
- RA versus DR sensitivity
- Bootstrap SIS intervals

```
fit_sens <- csdr(
  Y = Y, A = A, C = C,
  variants = c("RA", "DR"),
  L = 5, seed = 42,
  learners = learners,
  verbose = FALSE
)

summary(fit_sens)

outcome_fits <- nuisance_fits(fit, role = "outcome")
gps_fits <- nuisance_fits(fit, role = "gps")
```

Shared folds: Compare RA and DR dimensions, subspaces, and response curves.

Takeaways

When should I use CSDR?

Use CSDR when you want

- A **joint causal exposure–response**
- A **data-adaptive dimension**
- A flexible nonlinear response
- Exposure contribution summaries

CSDR does not replace

- Study design
- Confounder selection
- Positivity assessment
- Sensitivity analysis

One-sentence takeaway

CSDR learns the smallest exposure representation needed to preserve the causal mean response—and `csdr` turns that target into an applied workflow.

Paper and package

CSDR preprint

arxiv.org/abs/2606.14840

csdr R package

github.com/tXiao95/csdr

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Questions?

Appendix

Identification assumptions

1. **Consistency:** observed outcomes agree with potential outcomes under the observed exposure.
2. **Conditional exchangeability:** $Y^x \perp\!\!\!\perp X \mid C$.
3. **Positivity:** exposure values of interest are supported within levels of C.

Practical reminder

Dimension reduction may improve the support problem, but it does not justify extrapolation or repair an inadequate adjustment set.

Choosing among causal response constructions

Construction	Main nuisance inputs	Practical role
RA	Outcome regression	Lower-cost sensitivity analysis
DR	Outcome regression + GPS	Package default; robust to one correctly specified nuisance component
PO	Outcome regression + GPS	Doubly robust but noisier SDR response
RP	Models for $E(Y \mid C)$ and $E(A_j \mid C)$	Efficient under additive-confounding structure

Teach the DR workflow once; use RA as the clearest sensitivity analysis if time permits.